

### Project 3. Energy consumption modelling

Suppose you are a dynamic team of data analysts and project managers who are working for a research institute in the Sustainable Country. In this nation, there is a strong commitment to a greener, more sustainable future, and developing an effective energy consumption management stands as a top priority.

The ability to model energy consumption and use these models to estimate future energy needs is paramount. Underestimating demand could lead to devastating outages that impact lives and the nation's economy. On the other hand, overestimating could result in idle capacity, wasting valuable financial resources.

The objective of the project that your team is working on is to develop an energy demand model that can estimate and forecast future energy needs within the residential sector of the Sustainable Country. This model will serve as a vital tool for policy makers and energy-related organisations. To achieve this, your project employs multiple regression analysis. You will examine various factors that influence residential energy consumption, including resident population, GDP, household size, median household income, and the costs of residential electricity, etc. The developed model will not only facilitate a sustainable and reliable energy supply but also contribute to the nation's vision of a greener future.

Your tasks are the following:

- Use a country of your choice and think about it as it were the Sustainable Country. Give some explanation as to why you have selected this country.
- Identify the factors that influence the residential energy consumption. Use a literature search to find this out.
- Use secondary data sources to collect historical data about these independent variables along with the dependent variable for the model.
- Evaluate the nature of the relationship between each independent variable and the dependent variable.
- Construct a few different models and test that they satisfy the assumptions of the multiple regression model.
- Discuss the results of your analysis and compare them with the findings in the literature.
- Use the developed model to forecast the residential energy consumption for a five-year period.
- Comment on the limitations of the model and discuss the ways in which it could be improved.
- Provide recommendations for policymakers to effectively utilise the project's findings.

Before you embark on the project, you will need to work on the project outline. The guidelines are provided in a separate document.